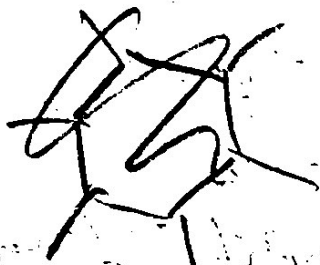


Call G a ~~5~~ column.



Apply eq + eq + eq

1) 2E - E + F = 2N

2) E + F = 2N

$$\frac{2E}{3} - E + F = 2N$$

$$\frac{1}{3}E + 2 = F$$

$$= 54E$$

$$\frac{1}{3}E + 2$$

$$I_N = 2 + E - N$$

$$\frac{3N}{2}$$

$$= \frac{1}{3} + \frac{2}{E}$$

⇒ Density

$$= \frac{1}{3}$$

$$N = 2 + E - F$$

$$\frac{3N}{2}$$

$$3N$$

$$2 + E - N$$

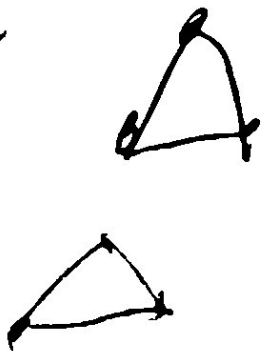
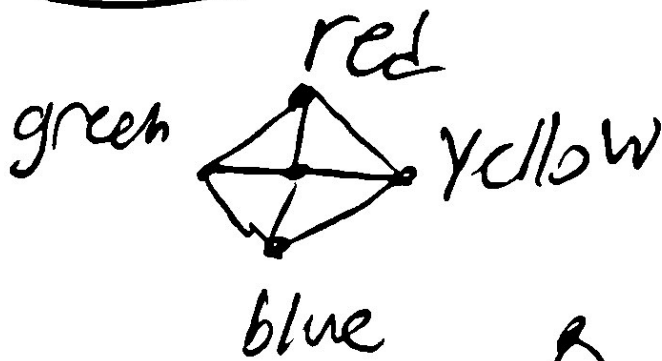
$$4 + 2E - N$$

$$= \frac{3(2 + E - F)}{4 + 2E - 2 - E + F}$$

$$4 + 2E - 2 - E + F$$

$$\begin{aligned}
 \text{arity} &= \frac{6 + 3E - 3F}{4 + E - 2 + F} \\
 &= \frac{6 + 3E - 3F}{2 + E + F}
 \end{aligned}$$

Kempe Chains!



if negative



Every group has only 1 subgroup  
 disjoint since identity  
 only has 1

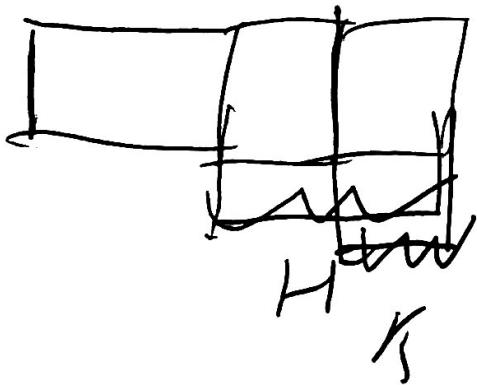
e.g. set of Integers is a group under addition

$(\mathbb{Z}, +)$

$\{0, -1, 1\}, +$

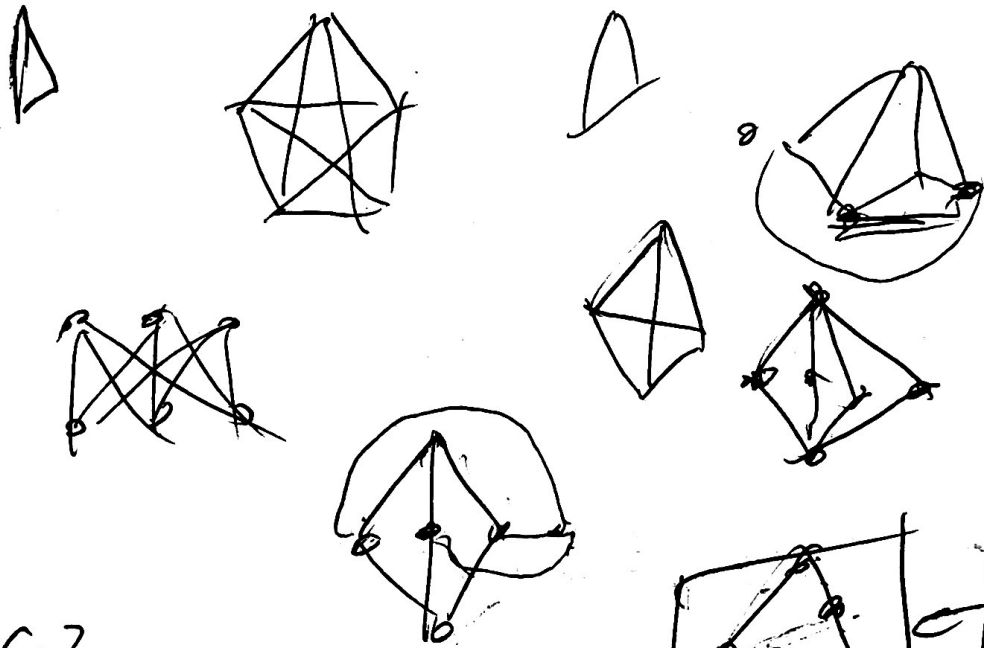
$H = \{50, -51, 51\}, +$

— no identity, no reverse for 50  
 ∴ not a subgroup



I think Lagrange's corollary says that  
 I only 1 Id, and a coset representative  
 operates on the identity, only exists for  
 1 coset representative Id on a subgroup  
 $H/K$  w/ the Identity.

$H \triangleleft G$ ,  $H$  is normal in  $G$ .



$$e \leq 3r$$

$$e \leq 3r - 6$$

$$r = e - v + 2$$

$$e \sum_{i=1}^r \deg(r_i) \geq 3r$$

all apply to planar graphs

$$2e =$$

$$e = \frac{\sum_{i=1}^r \deg(r_i)}{2}$$

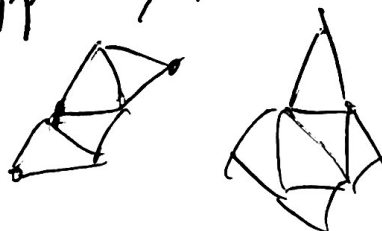
for all graphs

assume planar graph, Euler applies, then:

$$\frac{\sum_{i=1}^r \deg(r_i)}{3} \geq r$$

$$2e \geq 3r$$

$\therefore$  every edge used in 2 regions max



$$e \subseteq 3ub$$

for planar graphs

If  $H \triangleleft G$ , then the set of right cosets  $G/H$  form a group

$$[x] := \{y \in S \mid y \sim x\} \subseteq S$$

~~if  $x \in [y]$  then  $y \in [x]$~~

$$y \in [x] \Rightarrow (x, y) \in U \subseteq S \times S$$

$$\Rightarrow x \in [y] \quad \forall x, y \in U$$

$$\therefore [x] = [y] \text{ iff } y \in [x]$$

$$x \in S \Rightarrow x \in [y].$$

$$[x] = \{y \in S \mid x \sim y\} \subseteq S$$

equivalence classes

$$\text{if } x \equiv 0 \pmod{2}, [x] = \{s \in S \mid s \equiv 0 \pmod{2}\}$$

$$\text{if } x \equiv 1 \pmod{2}, [x] = \{s \in S \mid s \equiv 1 \pmod{2}\}$$

$\therefore$  partitioned by equivalence relation

An equivalence relation on a set is a partition if the partition gives rise to an equivalence relation whose equivalence class gives rise to the members of the partition

eg.  ~~$x, y \in \mathbb{Z}$~~

$$x \sim y \iff x, y \in \mathbb{Z} \mid$$

$$x \sim y \iff x, y \in \mathbb{Z} \mid x - y = 0 \pmod{5}$$

$$\text{symmetric: } (x, y) \in U \iff (y, x) \in U$$

$$\text{reflexive: } \forall x \in S \ (x, x) \in U$$

$$\text{transitive: } x, y, z \in S, (x, y) \in U \wedge$$

$$(y, z) \in U \Rightarrow (x, z) \in U$$

Congruent Relation:

$$\forall g_1, g_2, h_1, h_2 \in G:$$

$$g_1 \equiv g_2 \wedge h_1 \equiv h_2 \Rightarrow h_1 g_1 \equiv h_2 g_2$$

Q: if a set of  $N$  integers has  
<sup>unrestricted</sup>  
 many groups of 4 sum to  $N$   
 then, how many elements we  
 do have

$$1 \leq N \leq 200$$

$\rightarrow 200^4$  is max

$$200^4 = \frac{200!}{4! 146!}$$

how many elements  
 sum to  $N$ ?

if they are all distinct

assume  $a, b, c, d$  are  
 indices that sum to

$$N$$

$$a+b+c+d$$

$$2 + 2 + 2$$

$\in \mathbb{Z}$

$$N = N^2 + N^3 + N^4$$

$$= \frac{200 \cdot 199 \cdot 198 \cdot 197}{4 \cdot 3 \cdot 2}$$

$$= 25 \cdot 198 \cdot 66 \cdot 197$$

$$\leq 200^4$$

$$= 16 \cdot 120,000,000$$

$$= 1,600,000,000$$

$$= 1.6 \times 10^9$$

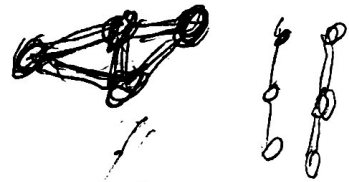
group of  $n$  : | 5 1 1 1 1

group of  $n$  :  $n^{n-1}$

$n=2, 3$

$$n^4 = 200^4 \approx 64 \text{ } 684400$$

first



$$\underline{200 \cdot 199 \cdot 198 \cdot 197}$$

197

197

$$= \frac{n^4}{4}$$

$$n \cdot (n-1) \cdot$$

$$n^4 \cdot n$$

What is TC?

find simpler way...

$$n! \cdot n \cdot n$$

$$n \cdot n \cdot (n^2)$$

• • • • •

$n-1$     1    4  
           2    3  
           3    2  
           4    1

$$n(n-1)$$

$n-1$     1  
 (1) (1)

1    3    (1x3 + 1x2 + 1x1)  
 2    2    (1x2 + 1x1)  
 3    1    (1x1)

$$n(n-1) + (n-2)(n-2) + (n-3)(n-3)$$

$$n-2 + n-3 + n-4$$

$$n^2 = \frac{n!}{(n-2)!}$$

$$= \frac{n(n-1)}{2}$$

$$n^2 = \frac{n(n-1)(n-2)(n-3)}{24}$$

$$n^2 = \frac{n(n-1)(n-2)}{6}$$

2 2 2 2  
2

111 333

11 33 101 3005

$$i, -i, 1$$

$$\{i, -1, -i, 1\}$$

$$\{ -1, 1 \} i \quad i \{ -1, 1 \}$$

$$= \{ -i, i \} \quad = \{ -i, i \}$$

$$= \{ -1, 1 \} -i = -i \{ 1, -1 \}$$

is  $\{ -i, i \}$  a normal group?

a subgroup  $H$  is normal if it is invariant under conjugation

for  $H \subset G$ ,  $H$  is normal i/f

$$gHg^{-1} = H \quad \forall g \in G$$

let  $H = \{ i, -i \}$

for a normal subgroup  $N$ ,

$G/N$  (quotient group)  $|N \triangleleft G$

$\Rightarrow G/N$  form a group.

$$G/N = aN = \{aN : a \in G\}$$

$$* N \triangleleft G, (aN)(bN) = (ab)N$$

Two pointers method

~~1 2 3 4 5 6~~

$\underbrace{1 \ 2 \ 3 \ 4 \ 5 \ 6}_n$

the  $n^{\text{th}}$  odd number is  $C(2n-1, 1) = 2n-1$

$$n(n+1)/2$$

$$4n+1$$

$$\frac{(4n+1)(4n)}{2}$$

$$n \frac{(4n+1)}{2}$$

~~1 2 3 4~~

=

1 2 3 4

$$n \frac{(4n+1)}{2} = n$$

(1 2 3 4)

$$N_0 = \frac{2n(2n+1)}{2}$$

$$N_{\text{odd}} = \frac{2n(2n+1)}{2} - 2n$$

~~1 2 3 4~~

$$\frac{n(2n+1)-n}{2}$$

$$(n+1)^2 - n^2 = 2n+1$$

$$a^2 + 2n + 1 - n^2 = 2n + 1$$

$$n^2$$

$$2^{n+1} - 1, n = n_0 \#$$

$$\Rightarrow 2^{n_0+1}$$

$$\cancel{4} (2^{n_0} - 1) - (2^{n+1} - 1)$$

$$\Rightarrow 4 \cdot 2^{n_0} - 1 - (2^{(n_0+1)} - 1)$$

$$= 2 \cdot 2^{n_0} + 0$$

$$= 2^{n_0+1}$$

$$2^0 + 2^1 + 2^2 + \dots + 2^n$$

$$= 1 + 2 + 4 + 8 + \dots + 2^n$$

~~Generally~~, induction requires a proposition

$$\log(2^0 + 2^1 + \dots + 2^n)$$

$$= \log 2^0 + \log 2^1 + \dots + \log 2^n$$

$$= \log 2 (1 + 2 + 3 + 4 + \dots + n)$$

$$\log_2 ab = \log_2 a + \log_2 b$$

$$\frac{a r^{n+1} - a}{r-1} = a \frac{r^{n+1} - 1}{r-1}$$

$$n(n^2-1)$$

$$= n(n+1)(n-1)$$

$$= n(n+1)(n-1) \pmod{3}$$

$$= (n \pmod{3})(n+1 \pmod{3})(n-1 \pmod{3}) \pmod{3}$$

assume  $n = 0$

$$0 \cdot 1 \cdot 2$$

$$n \pmod{3}$$

$$\begin{cases} 7^{n+2} - 8^{2n+1} \text{ divisible} \\ 5457 = 3 \times 1818 \end{cases}$$

Euclidean GCF algorithm

$$(a, b)$$

$$a \% b$$

$$a \% b = 0$$

$$\Rightarrow b = \text{GCF}(a, b)$$

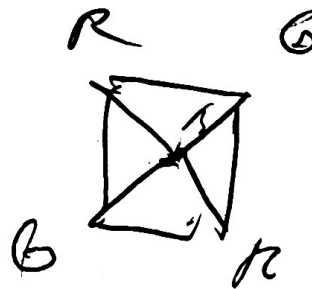
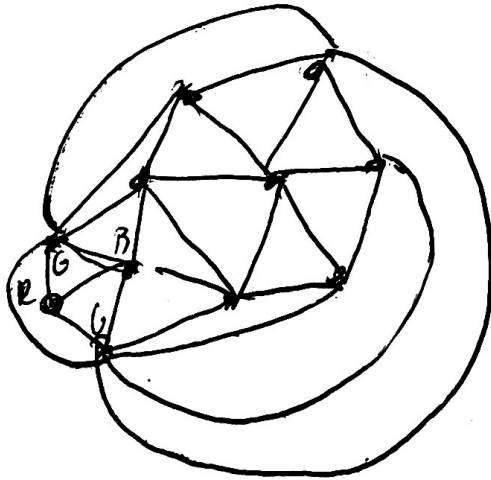
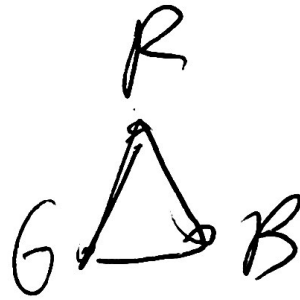
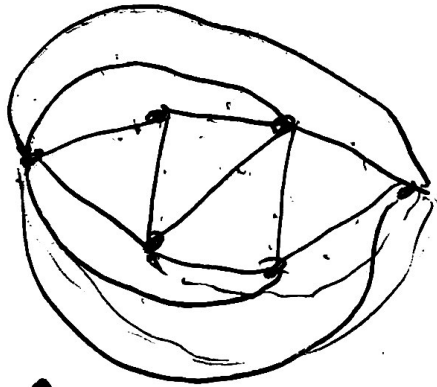
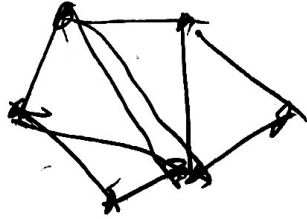
$$\hookrightarrow 16(32, 16)$$

$$32 \% 16 =$$

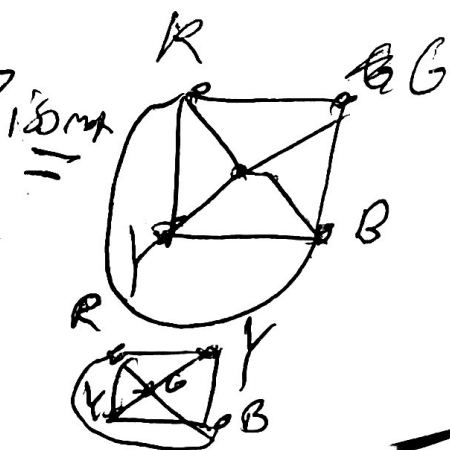
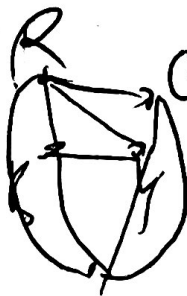
$$32 \% 16$$

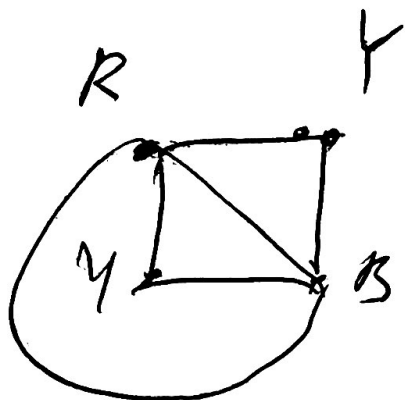
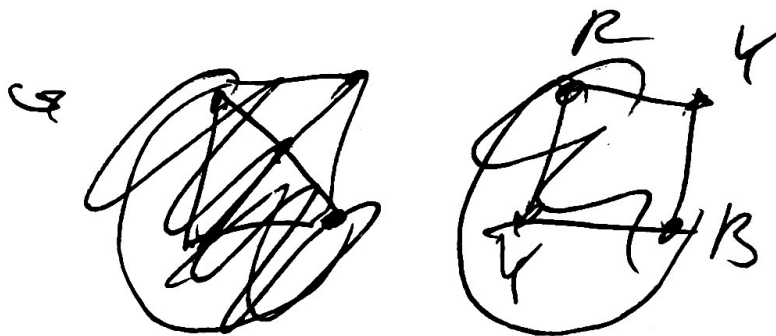
$$16 \mid 32 : 16 \% 16 = 0$$

induced from

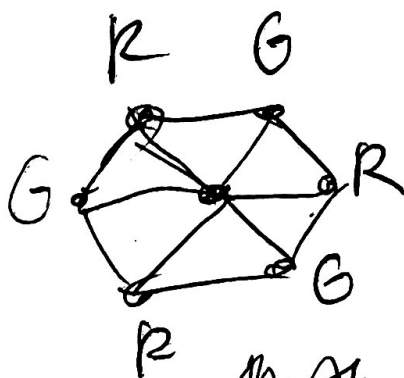
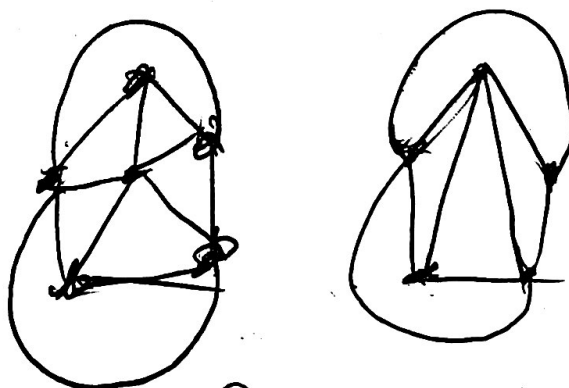


$$e = 3n - 6$$

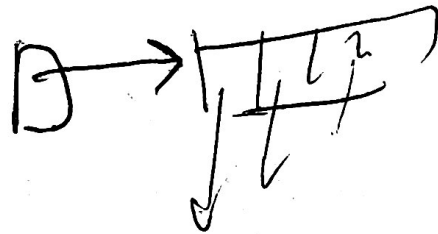
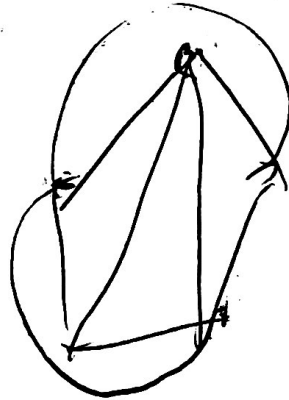
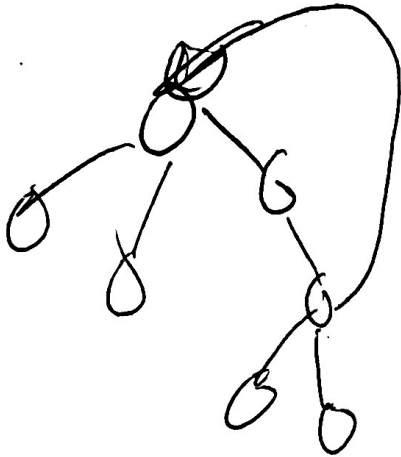




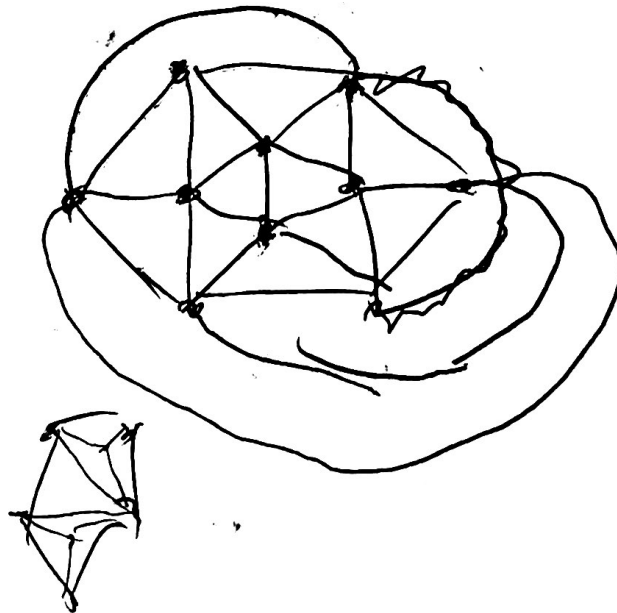
theorem: if a maximally planar graph removes a vertex and its edges and the remaining



$K_5$  - 1 edge



Birkhoff 1913,



No 633 configuration can appear in a minimal counterexample, if it did the minimal counterexample could be reduced

2 But every minimal counterexample has a configuration in the 633 list

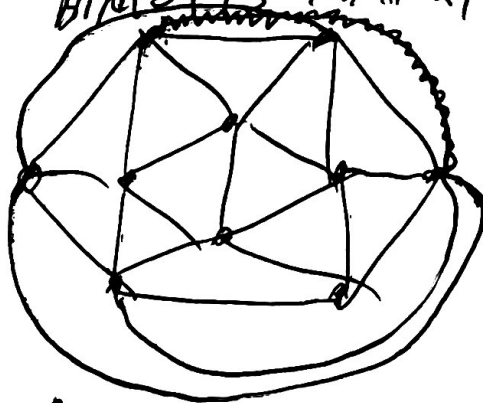
1) reducibility (reduced to what?)

2) unavoidability

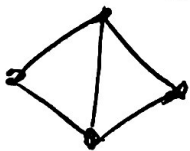
Example!

Internally 6 connected

Wirth's minimal counterexample

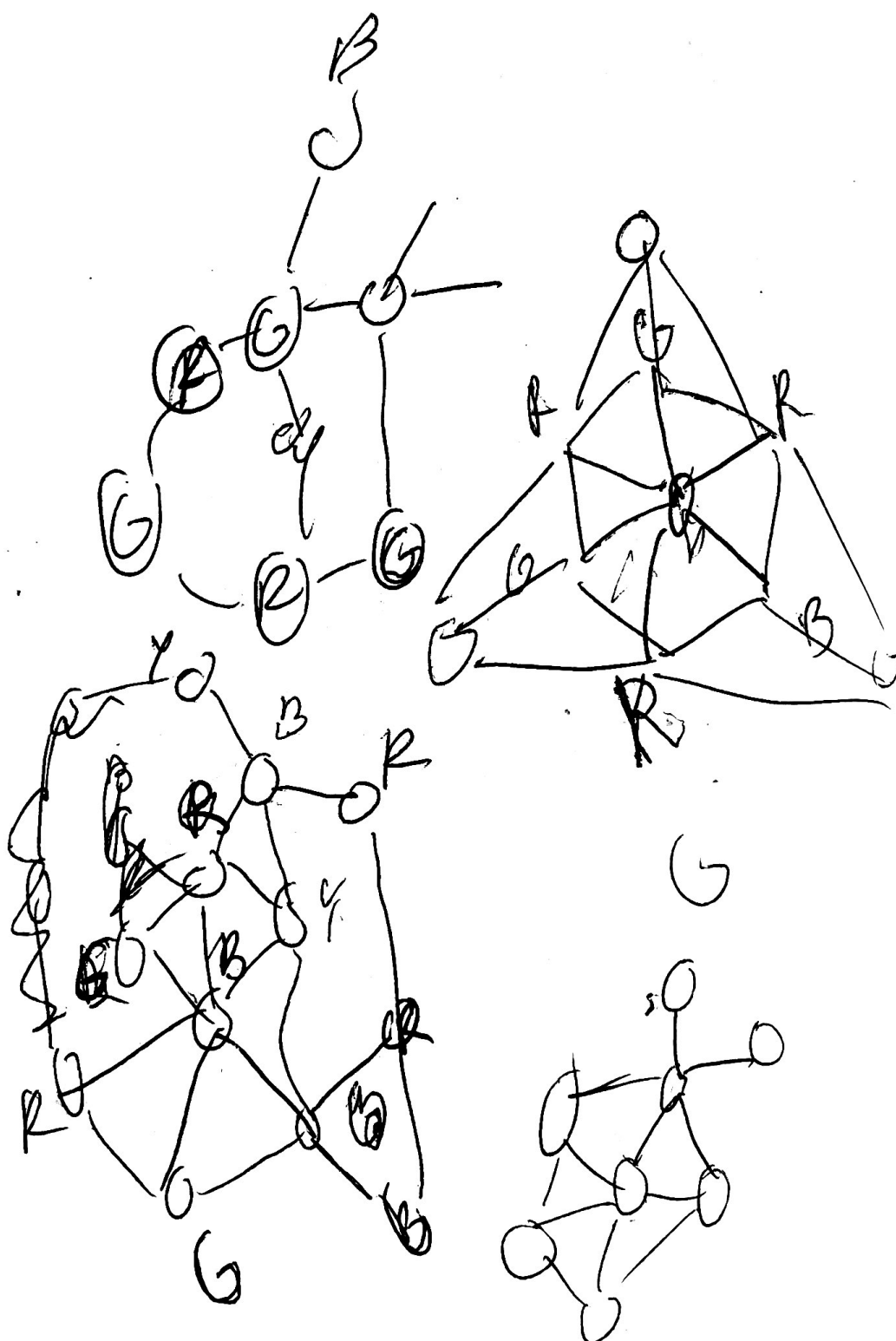


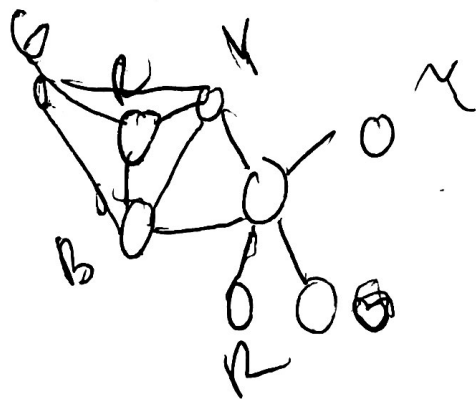
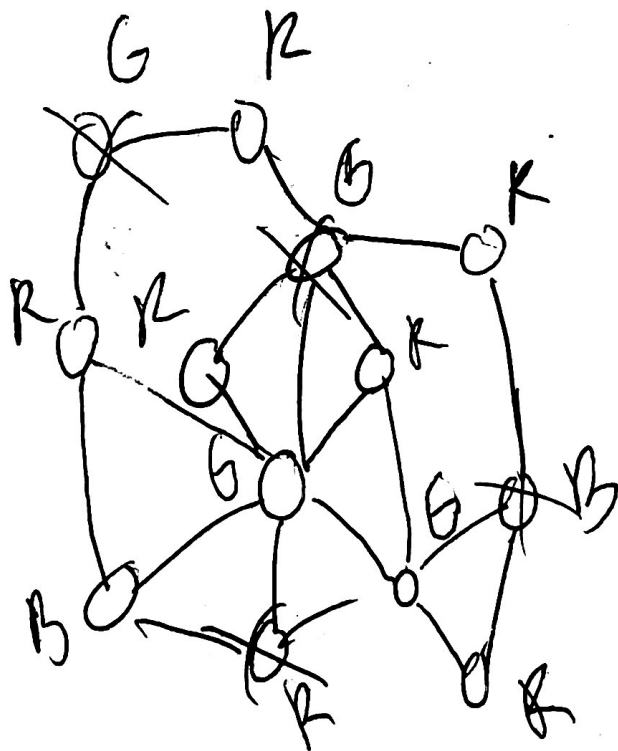
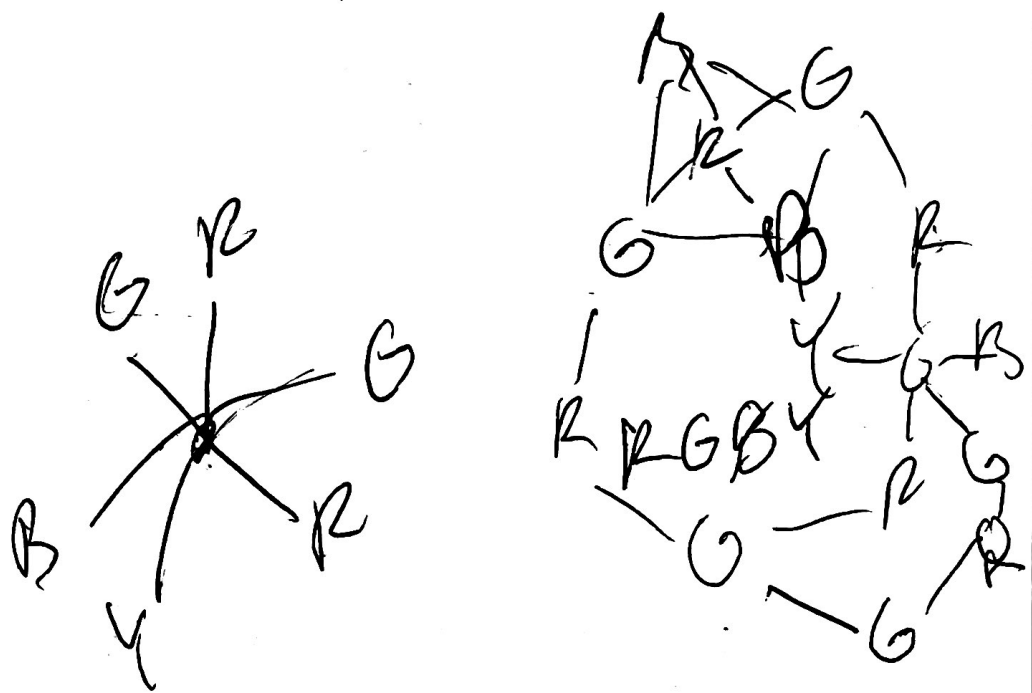
how can degree be 5, 6?



∈ The unavoidable set of reducible config.

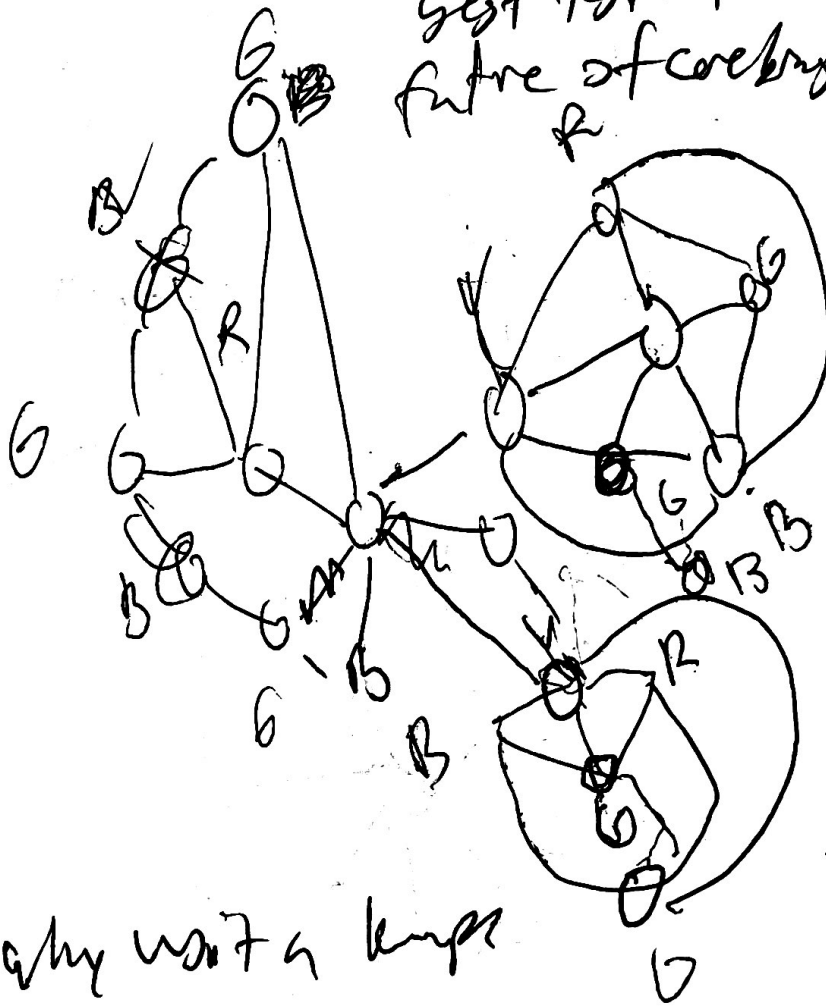
relates to Kempe?





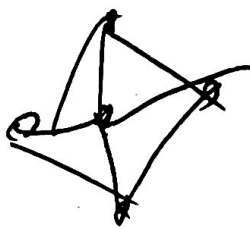
story?  
 best for?  
 future of core?  
 R

interns?

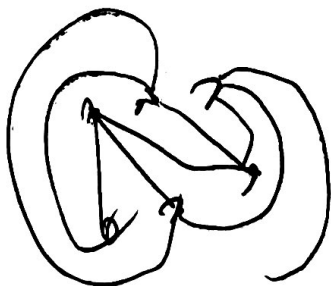
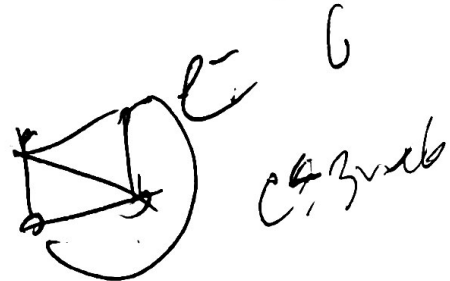


why wait a long?

then



3rd (124)

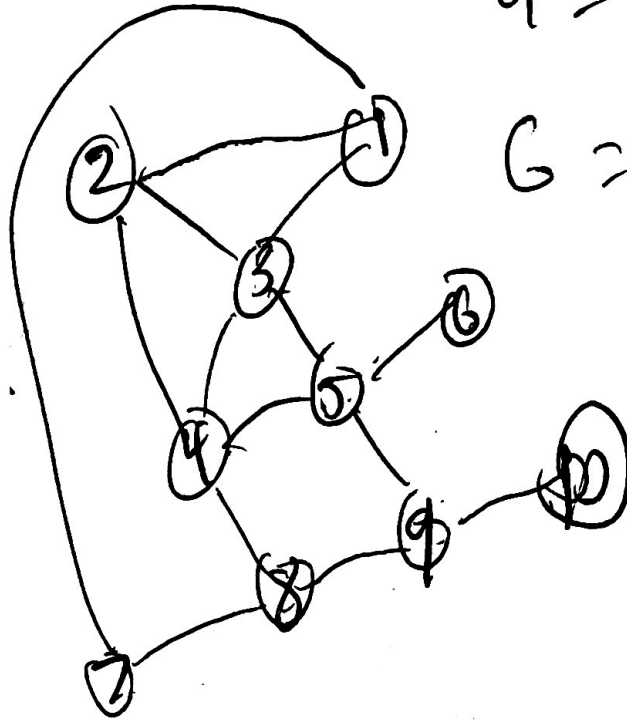


123  
 'or say don't be the  
 to the,

$$q = \frac{1}{2}bh$$

$$G = \frac{1}{2}b^4h^2$$

$$3 = h^2$$



$$\frac{1}{2}bh^2 \geq 9$$

$$\Rightarrow bh^2 \geq 18$$

$$\Rightarrow h^2 \geq \frac{18}{b}$$

$$\Rightarrow |h| \geq \sqrt{\frac{18}{b}}$$

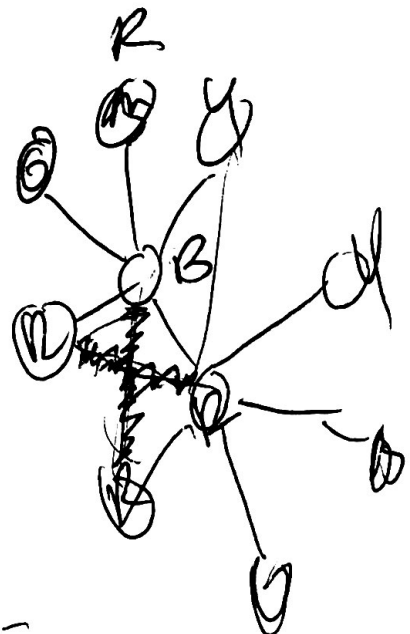
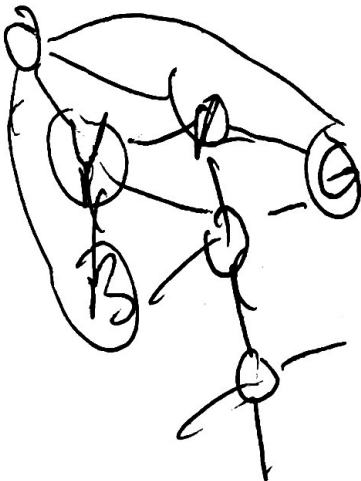
$$\frac{1}{2}bh \geq 9$$

$$bh \geq \frac{18}{b}$$

$$h \geq \frac{3}{b}$$

0000

6462



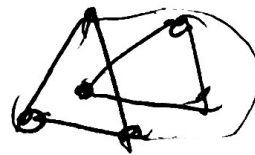
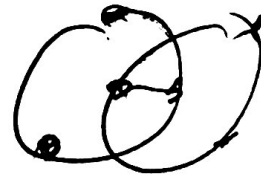
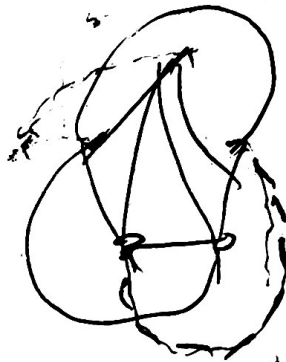
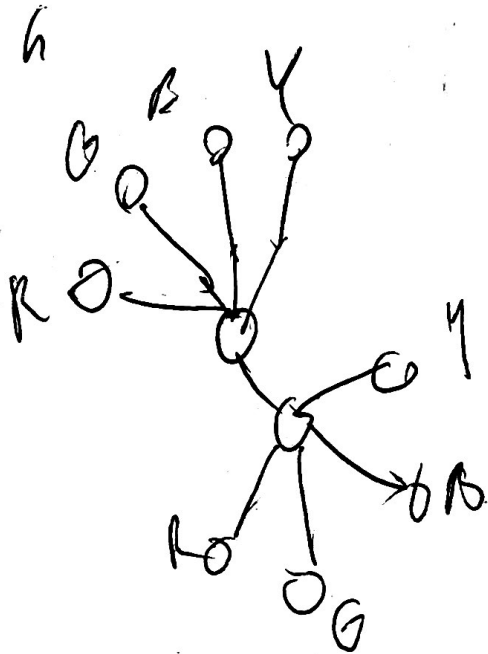
$$2^5 = 1 + 2 + 4 + 8 + 16$$

$$km = J$$

$$A = \begin{bmatrix} k \\ m \end{bmatrix}$$

$$\frac{km}{m^2}$$

$$= D1$$



1100  
2011  
8421

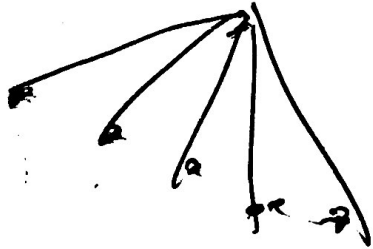
1011

1111

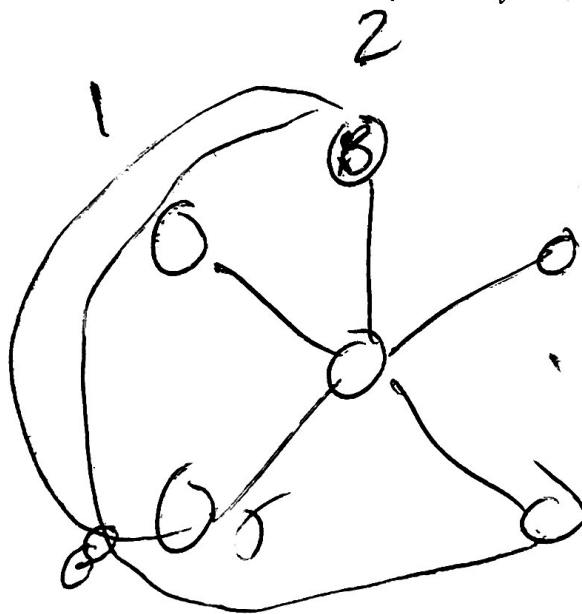
1011

6211

$$5.4 \quad \frac{6.5}{2} \quad 6^2 \quad \frac{6}{2} = \frac{6}{4.25}$$



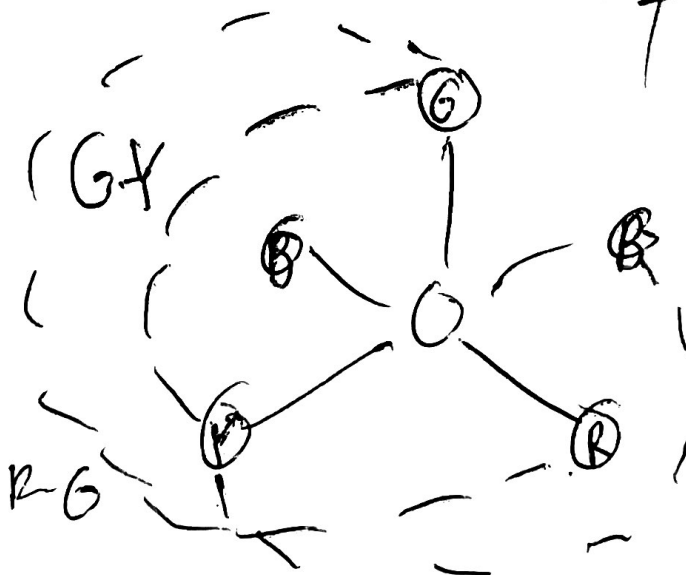
$$\frac{6.5}{2}$$



the

+

the





(R) - (R) - (G) - (G)

(R) - (R) - (G) - (R) - (G) - (G) - (G)

(R) - R - G - (G)

G toggle  
not possible

R not possible

(G) - (G)

G-Y???

(R) - R - G - R - (G)

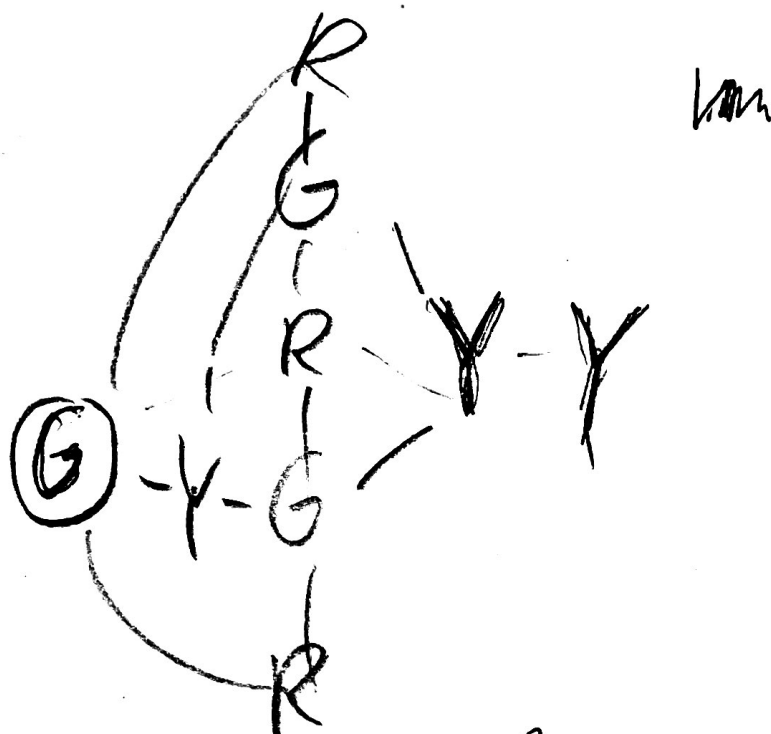
toggle

(R) - G - R - G - (R)

Can control end nodes colouring  
- assume 1st node doesn't change

(G) - (G)

(R) - G - G - G - G



2nd counterexamples

$$(R) - G - R - G - (G)$$

$$T_{ab} = T_{x_1, x_2} \cdots T_{x_{2k-1}, x_{2k}}$$

$$C_a = C(a+G)$$

$$= ca + cG$$

$$= ac + (b-b)c$$

$$= ac + bc - bc$$

$$= (ac - bc) + bc$$

$$= c(a-b) + bc$$

$$= bc$$

$$\Rightarrow c(a-b) = 0, \text{ but } c \neq 0$$

$$\Rightarrow a - b = 0$$

$$\Rightarrow a = b$$

Let  $m \in \mathbb{N}$ ,  $\forall a, b, a', b' \in \mathbb{Z}$

$$[a] = [a'], [b] = [b']$$

$$[a] = \{a_i \in \mathbb{Z}/m\mathbb{Z} \mid a_i \equiv a \pmod{m}\}$$

$$[b] = \{b_i \in \mathbb{Z}/m\mathbb{Z} \mid b_i \equiv b \pmod{m}\}$$

$$[a] + [b] = [a+b] = [a] + [b]$$

$$= [\{a_i + b_i\}]$$

$$= [c_i]$$

$$= [\{c_i \in \mathbb{Z}/m\mathbb{Z} \mid c_i \equiv x \pmod{m}\}]$$

$$\begin{array}{r} 55 \\ 17 \\ \hline 72 \end{array}$$

2

$$c \equiv a + b \pmod{m}$$

$$= a \pmod{m} + b \pmod{m}$$

$$= q_1 m + p + q_2 m + l$$

$$= m(q_1 + q_2) + p + l$$

$$\therefore [c] = \{ c' \in \mathbb{Z} \mid c = p + l \pmod{m} \}$$

$$c \in \mathbb{Z}/m\mathbb{Z},$$

$$= [a' + b'] \because [a] = [a']$$

$$[b] = [b']$$

$$(q_1 m + p)(q_2 m + l)$$

$$= q_1 q_2 m^2 + m(q_1 l + q_2 p)$$

$$+ pl$$

$$= m(q_1 q_2 m + q_1 l + q_2 p) + pl$$

$$\equiv pl \pmod{m}$$

$$ax \equiv 1 \pmod{m} \iff a \text{ and } m \text{ are coprime}$$

$$pl \equiv 1 \pmod{m}$$

$$\iff p, l < m \text{ are coprime}$$

$$\text{assume } p \text{ is not coprime } \exists l \text{ st. } pl \equiv m$$

$$\Rightarrow pl \equiv 0 \pmod{m}$$

If  $a, b$  are coprime,  $a < b$ , then

$$b = ax + r, \quad r > 0$$

$$\Rightarrow b - r = ax$$

$\Rightarrow$

$$ax + b - r = HCF(a, b)$$

assume  ~~$a, b$  are coprime~~  ~~$atb$~~   $hcf(a, b)$

$$\text{let } h_1 = gcf(a, b)$$

$$\Rightarrow \underline{a = k_1 h_1}$$

$$\Rightarrow h \frac{a}{h} + h \frac{b}{h} = a + b, \quad b > a$$

$$\text{or } \frac{hb}{h} - \frac{ha}{h} = h \frac{(b-a)}{h}$$

$$\text{assume } b-a = h$$

$$hcf(a, b) = h$$

$$ha - (hb) = A - B, \quad A = ha, \quad B = hb$$

$$\Rightarrow hb - ha = h(b-a) \quad B > A$$

$$3 \cdot 5 - 2 \cdot 5 = \quad b-a = 1$$

$$\Rightarrow b - (b-1) = 1 = hb - ha$$

$$\Rightarrow a = b-1$$

$$b = a+1$$

$$b-a \geq hK$$

$$\text{with } k \geq 1$$

$$hb_1 - ha_1 \geq hK, k \geq 1$$

$$hb \cdot$$

$$10 \cdot 30$$

$$\geq 35, 14 \neq 1$$

$$35, 15.$$

$$70, 15$$

$$105,$$

$$13 \neq 11 \cdot$$

$$13, 11, 13 \geq 143$$

$$5, 7, 3, 5$$

prove  $\text{gcd}(b, r) \geq 1$  if  $a, b$  coprime

$$a = p_1 p_2 \dots p_r$$

$$a < b$$

$$p_i \neq q_j$$

$$b = q_1 q_2 \dots q_s$$

$$p_i \neq q_j$$

let  $a$  be even

$$p \text{ not } (a) = 1$$

$$\begin{aligned} b \quad hb \quad B-A \\ &= hb - hA \\ &= h(b-a) \end{aligned}$$

$\bullet$   $b$  and  $a$  are coprime

$$\geq h(K), k \geq 0$$

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$$k = b - a$$

$$b - a$$

$$= h(b - a), h = 1$$

$$= 1 \cdot (b - a)$$

$$h \quad p_1 - q \quad \frac{p - q}{a p - b q} = 1$$

$$\Rightarrow a p = 1 + b q$$

$$p - q = 2(c)$$

$$\frac{(p - q)}{2} = c$$

assume  $a p - b q = 1$

$$a p = 1 + b q$$

$$p = 1 + q$$

any number is the sum of 1 + another

$$11 - 7 = 4$$

$$11 - 14 = -3$$

$$22 - 14 = 8$$

2 composite numbers

$$W \quad A = a_1, a_2, \dots$$

$$B = b_1, b_2, \dots$$

$a_i, b_i \in \mathbb{P}$

$$b - a$$

$$p - q$$

$$= h(22)$$

$$= (1)(11)$$

$$22 - 21$$

just any 2 composite numbers

$$22 - 21$$

$$2 \cdot 11 = 3 \cdot 7$$

$$22 - 21$$

- 2 numbers that differ by 1 are coprime
- any 2 coprime numbers can be made to differ by 1

an  $\#b \neq 1$  ;  $u, v$  coprime  
 assume not possible  $p_1, q_1$

$p_1 \quad q_1 \quad 3-5$

7-

Assume  $p - q \neq 1$

$$p = p_1 p_2 \dots p_r$$

$$q = q_1 q_2 \dots q_s$$

$p, q$  are coprime

$$\frac{p}{p_1} = p_2 p_3 \dots p_r$$

$$\frac{q}{q_1} = q_2 q_3 \dots q_s$$

$\frac{p}{p_1}$  &  $\frac{q}{q_1}$  are coprime

$a < b$  coprime

$$b-a = \phi$$

$$7, 17 \quad \phi = 10 > 1$$

$$14, 17 \quad \phi = 3 > 1$$

$$21, 17 \quad \phi = 4 > 1$$

$$28, 34$$

$$56, 34$$

$$7, 17$$

$$49, 68$$

$$h_c f(1)$$

$$17, 34, 51, 68$$

$$3, 6, 9$$

$$34$$

$$r =$$

$$p < q$$

$$p, q \text{ coprime}$$

$$0$$

$$q \equiv k \pmod{p}$$

$$p < q \quad \text{if } k = 1 \text{ or } p-1$$

$$p \mid k_i \equiv 1 \pmod{p}$$

solved

3.

assume a finite list of  $p$  numbers  
 $[0, p-1] \neq [0, p-1]$

can any number  $[1, p-1]$ , generate  
1 or  $p-1$ ?

$$\langle i \rangle_{\text{mod } p}$$

if  $p$  is even yes

if  $p$  is odd

$$\text{ex } \langle 3 \rangle_{\text{mod } 7}$$

$$\{3, 6, 2, 5, 1, 4, 6\}$$
$$3 \ 6 \ 9 \ 12 \ 15 \ 18 \ 21$$

$$9, 14$$

$$14 \equiv 5 \pmod{9}$$

$$\{5, 1$$

$$p, q, p < q$$

If any 2 coprime numbers, does ~~the~~

$\langle p \rangle_{\text{mod } p}$  generate a set that  
contains 1 or  $p-1$ ?